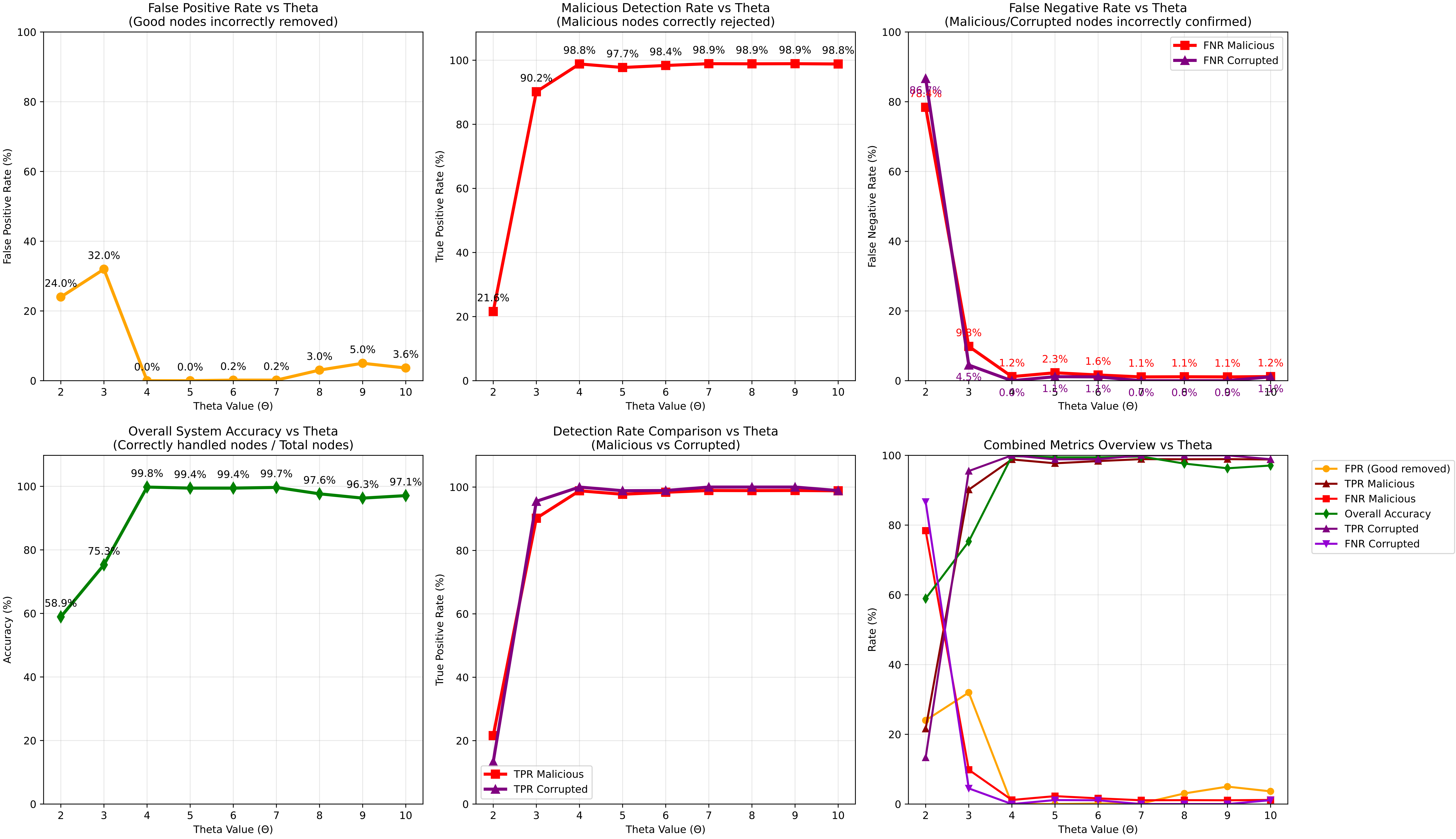
Theta Comparison - Confirmation Threshold Analysis



Theta Effectiveness Ranking

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THETA EFFECTIVENESS RANKING
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1. Theta = 4
   FPR: 0.0%, TPR: 98.8%, FNR: 1.2%, Acc: 99.8%
   Score: 197.4

2. Theta = 7
   FPR: 0.2%, TPR: 98.9%, FNR: 1.1%, Acc: 99.7%
   Score: 197.3

3. Theta = 6
   FPR: 0.2%, TPR: 98.4%, FNR: 1.6%, Acc: 99.4%
   Score: 196.0

4. Theta = 5
   FPR: 0.0%, TPR: 97.7%, FNR: 2.3%, Acc: 99.4%
   Score: 194.9

5. Theta = 8
   FPR: 3.0%, TPR: 98.9%, FNR: 1.1%, Acc: 97.6%
   Score: 192.4

6. Theta = 10
   FPR: 3.6%, TPR: 98.8%, FNR: 1.2%, Acc: 97.1%
   Score: 191.1

7. Theta = 9
   FPR: 5.0%, TPR: 98.9%, FNR: 1.1%, Acc: 96.3%
   Score: 189.1

8. Theta = 3
   FPR: 32.0%, TPR: 90.2%, FNR: 9.8%, Acc: 75.3%
   Score: 123.6

9. Theta = 2
   FPR: 24.0%, TPR: 21.6%, FNR: 78.4%, Acc: 58.9%
   Score: -21.9


THETA PARAMETER:
The Theta parameter controls the confirmation threshold
in the DAG structure. A node is confirmed when its
cumulative weight (CW) exceeds Theta.
Higher Theta values require more confirmations.

THETA VALUES TESTED:
Values: [2, 3, 4, 5, 6, 7, 8, 9, 10]
Configuration: Testing confirmation threshold effects
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Summary Statistics

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THETA COMPARISON ANALYSIS - SUMMARY
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Generated: 2025-08-31 18:56:44
Results Path: FL/results_theta

CONFIGURATION ANALYZED:
• Confirmation Threshold Testing
• Theta values tested: [2, 3, 4, 5, 6, 7, 8, 9, 10]

DETAILED RESULTS:
\n• Theta 2:\n  FPR=24.0%, TPR=21.6%, FNR=78.4%, Acc=58.9%\n\n• Theta 3:\n  FPR=32.0%, TPR=90.2%, FNR=9.8%, Acc=75.3%\n\n• Theta 4:\n  FPR=0.0%, TPR=98.8%, FNR=1.2%, Acc=99.8%\n\n• Theta 5:\n  FPR=0.0%, TPR=97.7%, FNR=2.3%, Acc=99.4%\n\n• Theta 6:\n  FPR=0.2%, TPR=98.4%, FNR=1.6%, Acc=99.4%\n\n• Theta 7:\n  FPR=0.2%, TPR=98.9%, FNR=1.1%, Acc=99.7%\n\n• Theta 8:\n  FPR=3.0%, TPR=98.9%, FNR=1.1%, Acc=97.6%\n\n• Theta 9:\n  FPR=5.0%, TPR=98.9%, FNR=1.1%, Acc=96.3%\n\n• Theta 10:\n  FPR=3.6%, TPR=98.8%, FNR=1.2%, Acc=97.1%\n

METRICS EXPLANATION:
• FPR (False Positive Rate): Good nodes incorrectly removed / Total good nodes sent
• TPR (True Positive Rate): Malicious nodes correctly detected / Total malicious nodes sent
• FNR (False Negative Rate): Malicious nodes incorrectly confirmed / Total malicious nodes sent
• Accuracy: Overall system accuracy (correctly handled nodes / total nodes)

CALCULATION METHOD:
• Each round selects K-1 = 9 nodes randomly from K = 10 peers
• Uses actual worker selection data from 3000_workers_selected.csv
• Sums Good_Nodes_Removed_This_Round and Malicious_Nodes_Confirmed_This_Round across ALL rounds
• Counts actual occurrences of each peer type in selected workers

OPTIMAL VALUES:
• FPR: 0% (no good nodes removed)
• TPR: 100% (all malicious nodes detected)
• FNR: 0% (no malicious nodes confirmed)
• Accuracy: 100% (perfect classification)

THETA PARAMETER:
The Theta parameter controls the confirmation threshold in the DAG.
A node is confirmed when its cumulative weight (CW) > Theta:
• Theta = 2: Low threshold (easier confirmation)
• Theta = 3: Default threshold
• Theta = 4: High threshold (stricter confirmation)
• Theta = 5: Very high threshold (very strict)
• Higher Theta values may improve security but reduce throughput
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